



 Roshan Shrestha

```
awscli user@ip-10-0-0-35 ~ % aws s3 cp /tmp/test.txt s3://network-vpc-project-roshan621
upload failed: .../tmp/test.txt to s3://network-vpc-project-roshan621/test.txt An error occurred (AccessDenied) when calling the PutObject operation: User: arn:aws:iam::3217568295:user/a3-cl-user is not authorized to perform 'PutObject' on resource: 'arn:aws:s3:::network-vpc-project-roshan621/test.txt' because no identity-based policy allows the s3:PutObject action
awscli user@ip-10-0-0-35 ~ %
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awscli user@ip-10-0-0-35 ~ %
```



Introducing Today's Project!

What is Amazon VPC?

Amazon VPC is a logically isolated space in the amazon cloudspace. It is useful because it helps us uniquely identify and isolate our resources from the rest in the cloudspace.

How I used Amazon VPC in this project

In today's project, I used Amazon VPC to create an EC2 instance. Then, I configured access keys to the EC2 instance CLI and interacted with my S3 bucket.

One thing I didn't expect in this project was...

On thing I didn't expect in this project was the numerous ways to configure access keys.

This project took me...

This project took me about 2 hrs.



In the first part of my project...

Step 1 - Architecture set up

In this step, we are creating a VPC from scratch and launching an EC2 instance inside it.

Step 2 - Connect to my EC2 instance

In this step, we are connecting directly to our EC2 instance.

Step 3 - Set up access keys

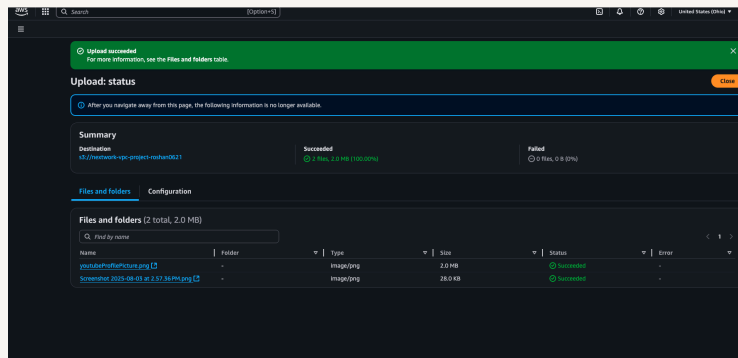
In this step, we are going to create access keys and give the EC2 instance access to our AWS environment.



Architecture set up

I started my project by launching a VPC and an EC2 instance in the public subnet of the VPC.

I also set up a general S3 bucket and uploaded two files into it.





Running CLI commands

AWS CLI is a software that you install and run on your computer to control AWS resources directly from your command line. I have access to AWS CLI because I am using the EC2 instance using EC2 connect and all EC2 instances come with it.

The first command I ran was "aws s3 ls". This command is used to list all S3 buckets you have access to.

The second command I ran was "aws configure". This command is used to configure user credentials to do things like accessing your S3 bucket list.

```
aws | [Search] [Option+S]

[ec2-user@ip-10-0-0-55 ~]$ aws s3 ls
Unable to locate credentials. You can configure credentials by running "aws configure".
[ec2-user@ip-10-0-0-55 ~]$
[ec2-user@ip-10-0-0-55 ~]$ aws configure
AWS Access Key ID [None]:
```



Access keys

Credentials

To setup my EC2 instance to interact with my AWS environment, I configured my credentials using the access keys generated earlier. Now, the CLI knows who I am and how to talk to AWS services.

Access keys are credentials for our applications and other servers to log into AWS and communicate with AWS services/resources.

Secret access keys are the password that pairs with your access key ID (username).
Need both to access AWS services.

Best practice

Although I'm using access keys in this project, a best practice alternative is to use AWS CloudShell or use the AWS CLI V2 and enable authentication through a user in IAM identity center.



In the second part of my project...

Step 4 - Set up an S3 bucket

In this step, we are launching a bucket in Amazon S3. We will try to access this from our EC2 instance.

Step 5 - Connecting to my S3 bucket

In this step, we are going to connect to the S3 bucket via our EC2 instance and interact with it.



Connecting to my S3 bucket

The first command I ran was "aws s3 ls". This command is used to list all S3 buckets you have access to.

When I ran the command "aws s3 ls" again, the terminal responded with the s3 bucket in my account. This indicated that my user credentials were setup correctly using the access key ID, secret access key, default region name, and default output format

```
[ec2-user@ip-10-0-0-55 ~]$  
[ec2-user@ip-10-0-0-55 ~]$  
[ec2-user@ip-10-0-0-55 ~]$ aws s3 ls  
2025-08-03 20:33:45 nextwork-vpc-project-roshan0621  
[ec2-user@ip-10-0-0-55 ~]$
```



Connecting to my S3 bucket

Another CLI command I ran was "aws s3 ls nextwork-vpc-project-roshan0621" which returned the list of the contents of my S3 bucket "nextwork-vpc-project-roshan0621".

```
[ec2-user@ip-10-0-0-55 ~]$ aws s3 ls
2025-08-03 20:33:45 nextwork-vpc-project-roshan0621
[ec2-user@ip-10-0-0-55 ~]$
[ec2-user@ip-10-0-0-55 ~]$
[ec2-user@ip-10-0-0-55 ~]$ aws s3 ls s3://nextwork-vpc-project-roshan-0621
An error occurred (NoSuchBucket) when calling the ListObjectsV2 operation: The specified bucket does not exist
[ec2-user@ip-10-0-0-55 ~]$ aws s3 ls nextwork-vpc-project-roshan0621
2025-08-03 20:35:16      28645 Screenshot 2025-08-03 at 2.57.36 PM.png
2025-08-03 20:35:16    2054525 youtubeProfilePicture.png
[ec2-user@ip-10-0-0-55 ~]$
[ec2-user@ip-10-0-0-55 ~]$
```



Uploading objects to S3

To upload a new file to my bucket, I first ran the command "sudo touch /tmp/test.txt". This command creates an empty file named test.txt in the /tmp/ directory.

The second command I ran was "aws s3 cp /tmp/test.txt s3://nextwork-vpc-project-roshan0621". This command will copy the test.txt file to my s3 bucket.

The third command I ran was "aws s3 ls s3://nextwork-vpc-project-roshan0621" which validated that the test.txt file was copied over to my s3 bucket.

```
ec2-user@ip-10-0-0-55 ~$  
ec2-user@ip-10-0-0-55 ~$  
ec2-user@ip-10-0-0-55 ~$  
ec2-user@ip-10-0-0-55 ~$ sudo touch /tmp/test.txt  
ec2-user@ip-10-0-0-55 ~$ ls /tmp/  
systemd-private-efaac9553824d8ab34e06f941947977-chromd.service-eflcs7      systemd-private-efaac9553824d8ab34e06f941947977-systemd-logind.service-fz10P  
systemd-private-efaac9553824d8ab34e06f941947977-dbus-broker.service-8Qvaw      systemd-private-efaac9553824d8ab34e06f941947977-systemd-resolved.service-ah19Dj  
systemd-private-efaac9553824d8ab34e06f941947977-policy-routesfens5.service-vq18Cx  test.txt  
ec2-user@ip-10-0-0-55 ~$ cat /tmp/test.txt  
ec2-user@ip-10-0-0-55 ~$  
ec2-user@ip-10-0-0-55 ~$ aws s3 cp /tmp/test.txt s3://nextwork-vpc-project-roshan0621  
upload failed: ../tmp/test.txt to s3://nextwork-vpc-project-roshan0621/test.txt An error occurred (AccessDenied) when calling the PutObject operation: User: arn:aws:iam::321775682595:user/s3-cli-user is not a  
uthorized to perform: s3:PutObject on resource: "arn:aws:s3:::nextwork-vpc-project-roshan0621/test.txt" because no identity-based policy allows the s3:PutObject action  
ec2-user@ip-10-0-0-55 ~$  
ec2-user@ip-10-0-0-55 ~$  
ec2-user@ip-10-0-0-55 ~$ aws s3 cp /tmp/test.txt nextwork-vpc-project-roshan0621  
Usage: aws s3 cp <LocalPath> <S3URI> or <S3URI> <LocalPath> or <S3URI> <S3URI>  
Error: Invalid argument type  
ec2-user@ip-10-0-0-55 ~$ aws s3 cp /tmp/test.txt s3://nextwork-vpc-project-roshan0621  
upload failed: ../tmp/test.txt to s3://nextwork-vpc-project-roshan0621/test.txt An error occurred (AccessDenied) when calling the PutObject operation: User: arn:aws:iam::321775682595:user/s3-cli-user is not a  
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ec2-user@ip-10-0-0-55 ~$
```



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